

# HcZZ ctag2d + JES/JER — Results

Production: combine\_run3\_100150\_ctag2d\_v8\_jecshifts

hplusc\_mva\_4class\_ctag2d\_scored\_v8 + hplusc\_mva\_4class\_ctag2d\_jecshifts\_scored\_v2,  
[100,150] GeV, no- $\bar{Z}X$  (MC-only), 15 bins (quantile + surgical rebin "2-4,7-10"),  
22 nuisances. Re-verified 2026-09-09 directly against the stored ROOT/JSON.

## Yields [100,150] GeV

| Process     | Rate      |
|-------------|-----------|
| Signal      | 0.053130  |
| ggZZ        | 2.941801  |
| qqZZ        | 36.318223 |
| Other_Higgs | 55.077609 |

S/ $\sqrt{B}$  = 0.0055

## Final limits (AsymptoticLimits, --run blind)

| Quantile            | r               | $\kappa C$   |
|---------------------|-----------------|--------------|
| 2.5%                | 196.3438        | 36.70        |
| 16%                 | 273.8452        | 50.15        |
| <b>50% (median)</b> | <b>412.0000</b> | <b>74.11</b> |
| 84%                 | 648.4839        | 115.09       |
| 97.5%               | 1004.9548       | 176.84       |

Without JES/JER (same production, JES/JER rows dropped):  $r=411.0/\kappa c=73.93$  —  
the 4 added nuisances shift the headline median  $\sim 0.2\%$ .

JES/JER's 4 rows (CMS\_scale\_j/CMS\_res\_j/CMS\_scale\_m/CMS\_res\_m) rank #6/#12/#15/#17  
by impact — individually bigger than several already-established nuisances  
(BR\_HZZ4l, pdf\_gg, scalevar\_muR, higgs\_plus\_c), even though adding all 4 barely  
moves the headline median. Likely explanation: CMS\_ctag2d already dominates the  
fit ( $\sim 10\times$  the #2 nuisance), and JES/JER's shape is largely degenerate with/  
absorbed by it — plausible since JES shifts change which jets get c-tagged.  
Not independently verified via a correlation matrix.

Systematics: type, assigned value, and impact

Ranked by impact\_r (combineTool.py -M Impacts, --expect-signal 412.0). InN rows show the assigned size per affected process; shape rows have no single size, so the value column shows the post-fit constraint instead (all  $\approx$  prefit  $\pm 1\sigma$  here — expected for a blind Asimov fit, not a sign of anything wrong).

| #  | Systematic             | Type            | Value (InN size / pull)    | Impact |
|----|------------------------|-----------------|----------------------------|--------|
| 1  | CMS_ctag2d             | shape           | pull -0.00 (-0.501/+0.443) | 180.46 |
| 2  | QCDscale_qqZZ          | InN             | 1.04 (qqZZ)                | 17.57  |
| 3  | ps_isr                 | shape           | pull +0.00 (-0.992/+0.993) | 17.19  |
| 4  | scalevar_muF           | shape           | pull +0.00 (-0.990/+0.993) | 16.15  |
| 5  | QCDscale_gg            | InN             | 1.039 (ggZZ, Signal)       | 14.32  |
| 6  | CMS_scale_j            | shape (JES/JER) | pull -0.00 (-0.985/+0.985) | 13.64  |
| 7  | BR_HZZ4l               | InN             | 1.02 (Signal, Other_Higgs) | 12.64  |
| 8  | CMS_eff_e_reco_below20 | shape           | pull -0.00 (-0.987/+0.987) | 10.63  |
| 9  | pdf_gg                 | InN             | 1.032 (ggZZ, Signal)       | 9.34   |
| 10 | scalevar_muR           | shape           | pull +0.00 (-0.993/+0.994) | 8.84   |
| 11 | higgs_plus_c           | shape           | pull -0.00 (-0.990/+0.991) | 8.50   |
| 12 | CMS_res_j              | shape (JES/JER) | pull -0.00 (-0.993/+0.992) | 6.01   |
| 13 | ps_fsr                 | shape           | pull +0.00 (-0.991/+0.994) | 3.43   |
| 14 | lumi_Run3              | InN             | 1.014 (all processes)      | 3.30   |
| 15 | CMS_scale_m            | shape (JES/JER) | pull +0.00 (-0.992/+0.992) | 2.67   |
| 16 | kfactor_ggZZ           | InN             | 1.10 (ggZZ)                | 2.63   |
| 17 | CMS_res_m              | shape (JES/JER) | pull +0.00 (-0.992/+0.992) | 2.50   |
| 18 | CMS_eff_e_reco_20to75  | shape           | pull -0.00 (-0.994/+0.994) | 2.23   |
| 19 | CMS_pileup             | shape           | pull +0.00 (-0.994/+0.993) | 1.79   |
| 20 | lhe_pdf                | shape           | pull -0.00 (-0.994/+0.994) | 0.35   |
| 21 | CMS_eff_e_reco_above75 | shape           | pull +0.00 (-0.994/+0.994) | 0.12   |
| 22 | CMS_eff_m_id           | shape           | pull +0.00 (-0.994/+0.994) | 0.02   |

